110 kV Transmission Network Technical Analysis

abstract

This repository contains the technical report and simulation results for a 110 kV transmission network analysis conducted using DIgSILENT PowerFactory. The study explores the progression of a high-voltage loop system from a stable baseline to a critical overload state, followed by various mitigation strategies using local generation.

1 Project Overview

The project models a high-voltage backbone consisting of five primary substations interconnected in a closed-loop configuration. The analysis evaluates system performance across four distinct operational scenarios to determine the most effective method for managing thermal congestion and maintaining voltage stability.

2 System Specifications

- **Voltage Level:** 110 kV.
- **Grid Reference:** A synchronous machine at the main node serves as the slack bus (1.0 pu voltage).
- **Infrastructure:** Mix of overhead lines and underground cables with specific impedance and thermal profiles.
- **Demand:** Distributed PQ and $P, \cos\phi$ loads simulating industrial and residential profiles.

3 Simulation Scenarios

3.1 Initial State (Baseline)

Establishing a stable healthy baseline where all elements operate within design parameters.

- **Voltage:** Stable profile between 1.00 and 1.01 pu across all nodes.
- **Loading:** Line 5 experiences the highest loading at 76.5%.

3.2 System Overload

A significant increase in power demand to simulate peak grid stress.

- **Thermal Violation:** Line 5 becomes severely overloaded at 113% capacity.
- **Voltage Degradation:** Significant drops observed, with Substation 3 falling to 0.97 pu.

3.3 Mitigation Phase A: Inductive Generation

Integration of a secondary synchronous machine at the Substation/BB operating in Inductive mode.

- **Results:** Successfully reduced Line 5 loading to 84.4%.
- **Impact:** Redistributed current flows, increasing stress on Line 4 to 60.2%.

3.4 Mitigation Phase B: Capacitive Generation

Transitioning the secondary generator to Capacitive mode to inject reactive power locally.

- **Results:** Superior thermal relief, reducing Line 5 loading further to 64.5%.
- **Efficiency:** Reduced the reactive power burden on the slack generator to 60.10 Mvar.

4 Key Findings

- **Strategic Local Generation:** Essential for relieving primary transmission paths during peak stress.
- **Capacitive vs. Inductive:** Capacitive mode proved significantly more efficient, reducing peak loading on critical lines by nearly 50%.
- **Voltage Stability:** Local capacitive injection provides a more robust solution for current-limited branches while maintaining a stable voltage range of 0.98–1.01 pu.

Summary of Line Loading Results

Table 1: Comparison of Line 5 loading across scenarios

Scenario	Line 5 Loading (%)
Baseline	76.5%
System Overload	113%
Mitigation A (Inductive Gen)	84.4%
Mitigation B (Capacitive Gen)	64.5%